Python Docstrings

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Docstrings

- Strings used for code documentation
- Enable automatic generation of docs in different formats (HTML, etc.)
- Different formats for Python
  - numpydoc
  - ...

Required Packages

- In our practical sessions, we will use
  - **sphinx**: to automatically generate the documentation from code docstrings
  - **numpydoc**: as our standard to format our docstrings
    - https://numpydoc.readthedocs.io/en/latest/
  - **sphinx_rtd_theme:** HTML popular theme for docs

Writing Docstrings with Numpydoc

- A numpydoc docstring is a triple-quoted string right below the def line, made of named sections underlined by dashes
  - **Summary**: one line, imperative mood, ending with a full stop
  - **Parameters**: one entry per argument, name : type on the first line, description indented below
  - **Returns**: type and meaning of the returned value; name it only if useful
  - **Raises**: each exception the function raises on purpose
  - **Examples**: doctest-style usage, >>> call followed by the expected output
- Sections appear in this order, and sections with nothing to say are left out
  - Optional arguments are written as name : type, optional
  - Sphinx reads these sections and renders them as the API documentation

Numpydoc: An Example

```
def load_dataset(path, normalize=True):
    """Load a dataset from a CSV file.

    Parameters
    ----------
    path : str
        Path to the CSV file.
    normalize : bool, optional
        If True, scale features in [0, 1]. Default True.

    Returns
    -------
    x : ndarray
        Feature matrix, shape (n_samples, n_features).
    y : ndarray
        Labels, shape (n_samples,).

    Examples
    --------
    >>> x, y = load_dataset("data.csv")
    >>> x.shape
    (100, 4)
    """
```

Doctest: Examples as Executable Tests

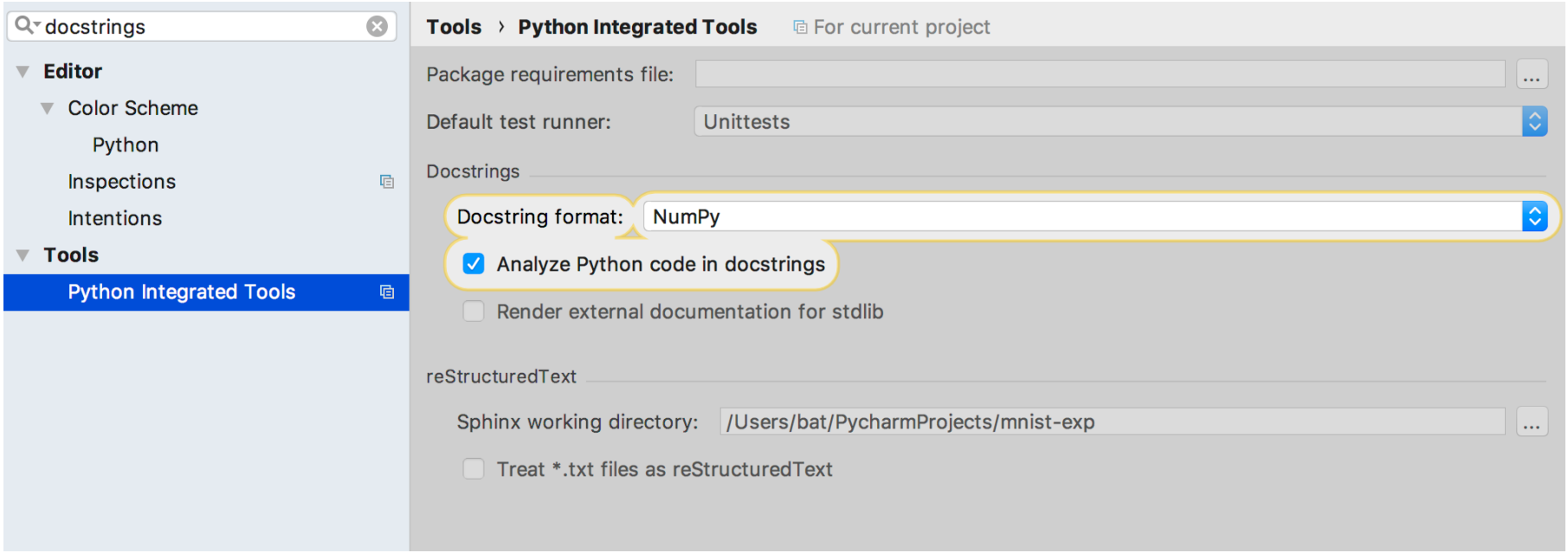
- Every **>>>** line in Examples is run as a **doctest**: the text under it is the output Python must produce
  - pytest --doctest-modules dataset.py, or python -m doctest -v dataset.py
- The comparison is on the printed text, character by character, so a different shape, a different order or an extra space fails the test
  - Keep examples small and deterministic: no random seeds, no file paths, no timestamps

```
$ pytest --doctest-modules dataset.py

File "dataset.py", line 20, in dataset.load_dataset
Failed example:
    >>> x.shape
Expected:
    (100, 4)
Got:
    (100, 5)
1 items had failures:  *** FAILED ***
```

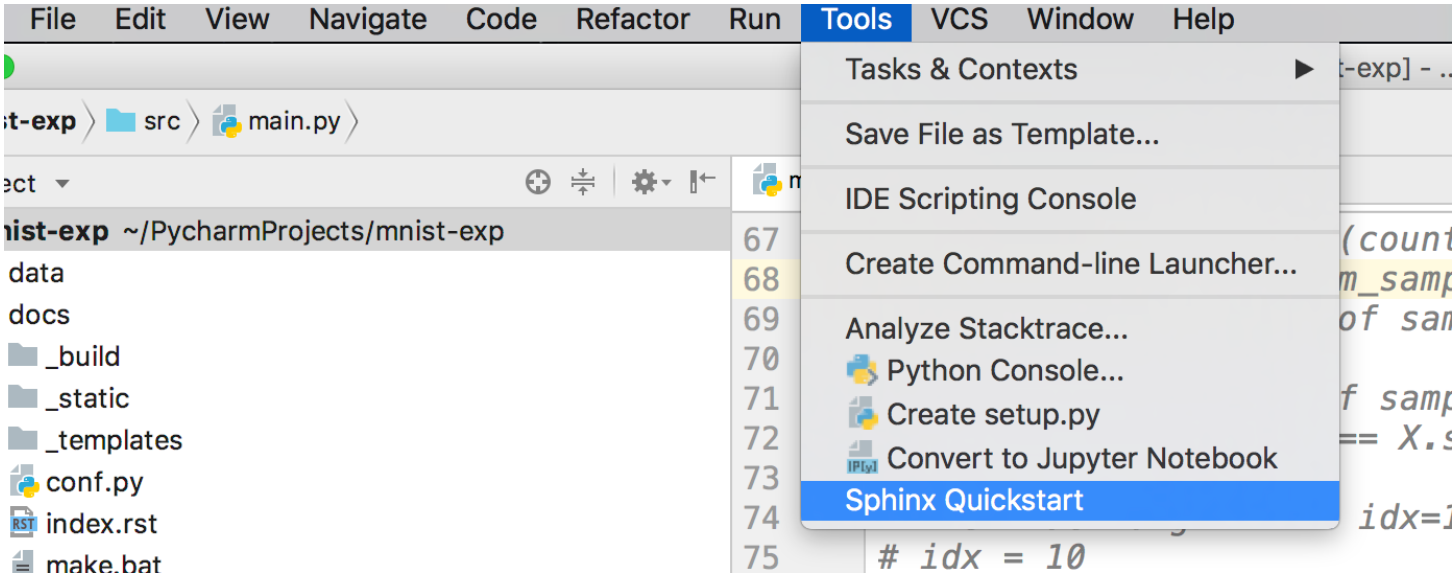
Configuring PyCharm

- Step 1: Set numpydoc as the default docstring format



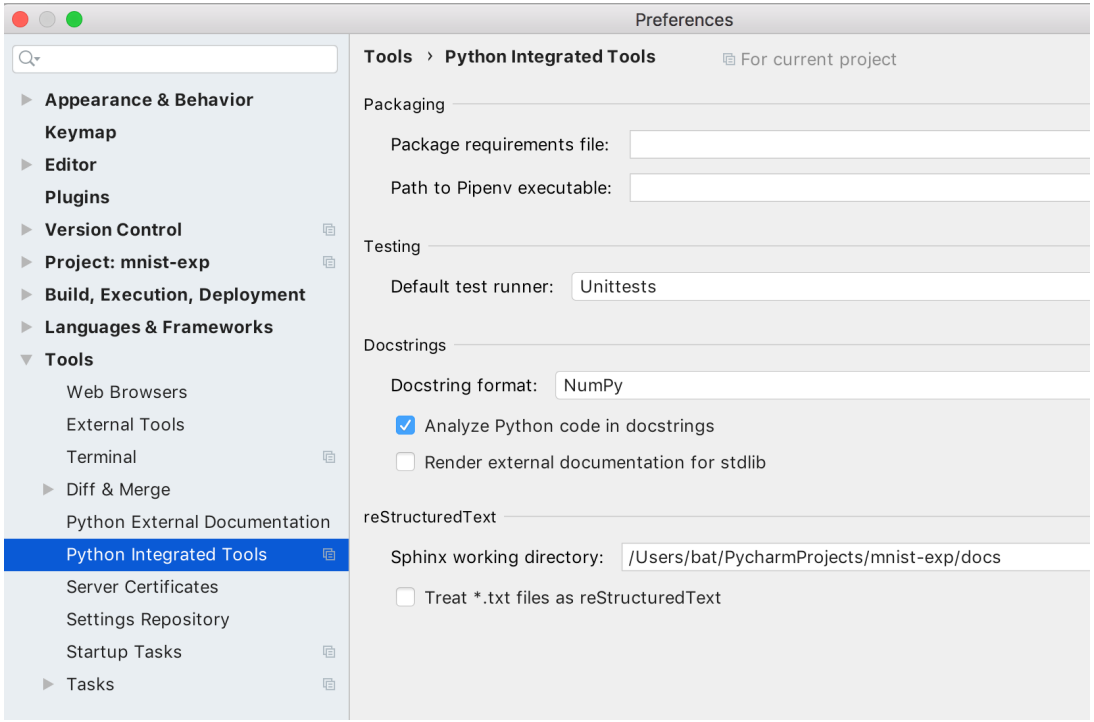
Configuring PyCharm

- Step 2: Set-up Sphinx configuration



Sphinx Quickstart

- Automatically configures sphinx environment
  - Set sphinx working directory «docs»
- Suggested preferences:
  - Do not separate source and build



Sphinx Quickstart

- The summary of your configuration is stored in **conf.py**
  - Set path to your src folder

```
# -- Path setup --------------
import os
import sys
sys.path.insert(0, os.path.abspath('../src/'))
```

  - To use the sphinx rtd theme, set:

```
html_theme = 'sphinx_rtd_theme'
```

  - Add autodoc extension:

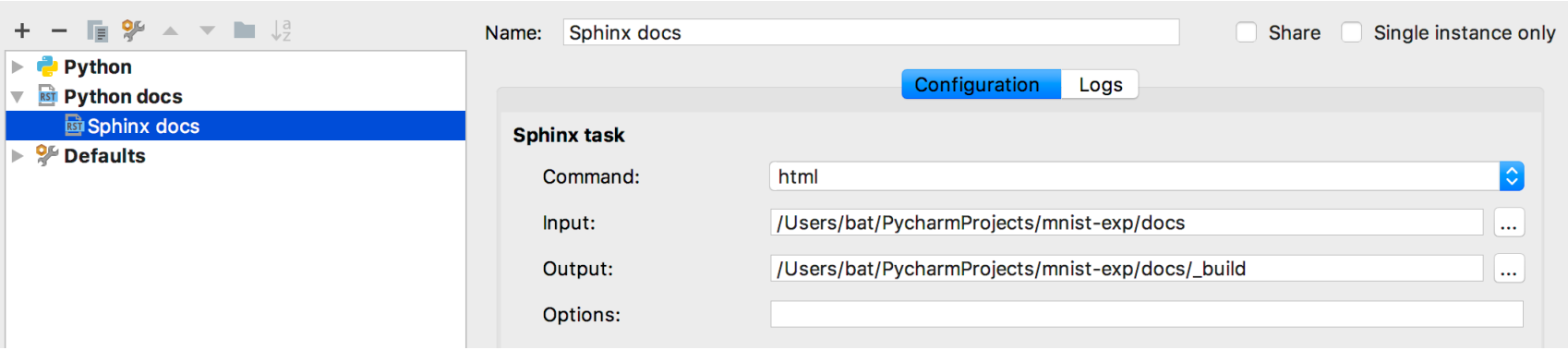
```
extensions = ['sphinx.ext.autodoc']
```

- **Index.rst** contains the name of the modules to be included
  - To add ours, let's just add the following commands into that file

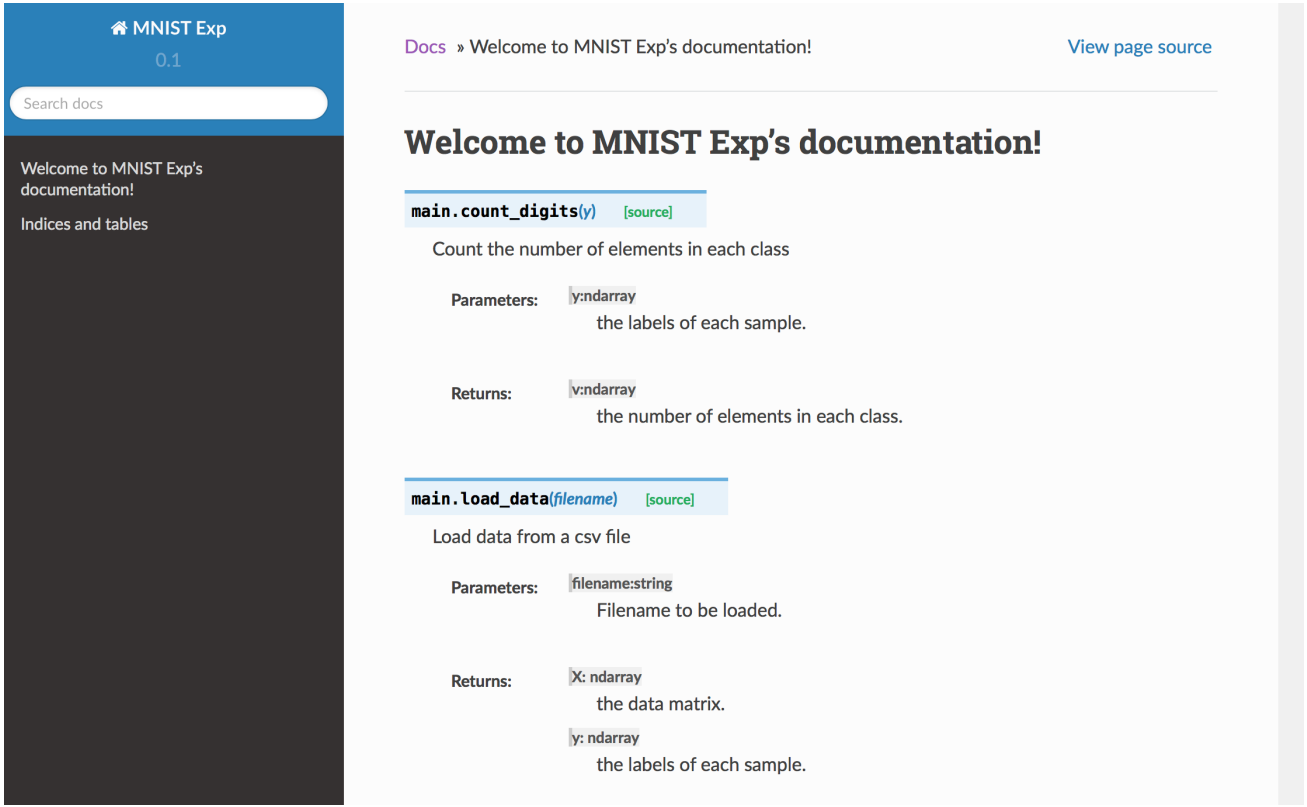
```
.. automodule:: your_file_or_package_name
   :members:
   :undoc-members:
```

Run Configuration for Sphinx docs in PyCharm

- Run → Edit Configurations
- Create a configuration 'Sphinx docs' from the defaults
- Set Input and Output folder
  - Output is just input_folder/_build



Finally...


MNIST Exp
0.1
Search docs
Welcome to MNIST Exp's documentation!
Indices and tables
Docs » Welcome to MNIST Exp's documentation!
View page source
Welcome to MNIST Exp's documentation!
main.count_digits(y) [source]
Count the number of elements in each class
Parameters: y:ndarray
the labels of each sample.
Returns: v:ndarray
the number of elements in each class.
main.load_data(filename) [source]
Load data from a csv file
Parameters: filename:string
Filename to be loaded.
Returns: X: ndarray
the data matrix.
y: ndarray
the labels of each sample.

See also...

- Blog post by Brendan Hasz
  - https://brendanhasz.github.io/2019/01/05/sphinx.html